Coordinates and inequalities

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: Use coordinates to locate and reflect points, and graph sets of solutions.

R1. Find $1.4 \div 0.07$.

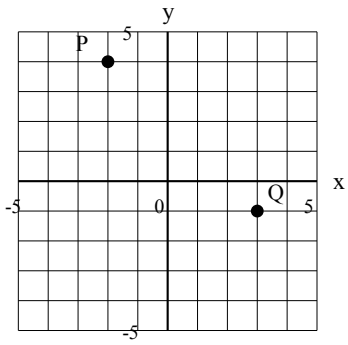
R2. Solve $(\frac{2}{5})\, x = 10$.

R3. Find the surface area of a closed 2 by 3 by 5 cm box.

R4. Find $10 - 2.286$.

P1. Plot A(2, 3), B(−4, 1), C(−3, −2), and D(1, −4). Name the quadrant of each.

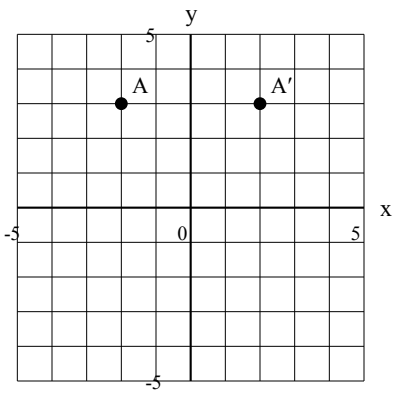
P2. Read P and Q from the grid. State the horizontal move first.



P3. Reflect (−2, 3) across the x-axis. Then reflect the original point across the y-axis.

P4. Find the distance from (−2, 3) to (−2, −4), and from (−2, 3) to (5, 3).

Lesson model



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Practice and compare methods

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P5. A rectangle has vertices (−3, 2), (2, 2), (2, −1), and (−3, −1). Find its perimeter.

P6. Graph $x > 2$ and $x \geq 2$ on separate number lines. Is 2 a solution of each?

P7. A ride requires a height of at least 48 inches. Write an inequality for h. Is 48 included?

F. Reflect (−5, −2) across the y-axis: (__, __). The distance between the points is __.

D. Nia writes $x > 6$ for “at least 6.” She leaves 6 out. Explain and repair her graph.

M1. Find 1.45 + 3.078.

M2. Find $3n + 1$ when $n = 4$.

M3. A rectangle is 8 cm by 2 cm. Find its area.

Worked example - compare with the slide model

Find the distance from (−1, −3) to (−1, 2).

The x-coordinate stays −1, so the segment is vertical.

From −3 to 0 is 3 units. From 0 to 2 is 2 units.

Total distance = 3 + 2 = 5 units.

Result: 5 units

Coordinates and inequalities

Challenge, explain and check

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C1. A delivery route goes from A(−4, 2) to B(3, 2), then to C(3, −3). Find the route length. Reflect all three points across the y-axis and give the new route length. Explain why reflection preserves the distance.

C2. A club accepts whole-number ages of at least 10 but less than 14. A second activity is for ages at most 12. List the ages allowed in both. Write each inequality and explain whether ages 10, 12, and 14 are included.

Finish the lesson

S. A reflection across the y-axis changes __ and keeps __.

E. Reflect A(−4, 1.5) across the y-axis. Give the image and the distance from A to its image.